

Exp No:28

COUNTING SEMAPHORE IN SYSTEMC

AIM:

To implement Counting Semaphore allowing multiple resource access.

PROGRAM:

```
#include <systemc.h>

SC_MODULE(counting_semaphore) {

    int sem;

    void task() {
        while(true) {
            wait(1, SC_SEC);

            if(sem > 0) {
                sem--;
                cout << "Task Accessing Resource at "
                    << sc_time_stamp()
                    << " Remaining: " << sem << endl;

                wait(2, SC_SEC);
                sem++;
            }
        }
    }

    SC_CTOR(counting_semaphore) {
```

```

sem = 2;

SC_THREAD(task);

SC_THREAD(task);

SC_THREAD(task);

}

};

int sc_main(int argc, char* argv[]) {
    counting_semaphore obj("Counting");
    sc_start(10, SC_SEC);
    return 0;
}

```

